

New Type LED Chip Mounter

SIGPlace

Operation Manual



Caution

Thank you for purchasing our products, for using our LED mounter safely, please read thoroughly and understand fully before operating the machine. Keep the manual in good condition for any further reference.

- (1) Without permission, anyone is strictly prohibited to copy or duplicate arbitrarily for parts of or the whole content of this manual (including software and program)
- (2) The contents of this book are subject to change without prior notice.
- (3) We strived to make it 100% when we were writing this manual. In case there is some error, omission or doubtful content, please contact with the distributor or our company.
- (4) Declaration: Our company is no responsible for any accidents caused by faulty operation, no matter it is misled by article (3) or not

In order not to get injured, any operator who is actually operating this SMT machine and its supporting device (hereafter referred to as machine) or any maintenance personnel who to maintain and repair this machine should read below safety precautions thoroughly and fully before operating the machine.

In the “safety precautions” of the manual, there is some other information not just the specification of the product purchased by the customer.

In addition, in order to make you understand the warning sign of the manual and product fully, we classify them as below. Please understand them fully and comply with them.

1)The degree of warning

 Danger	It means if the operator or a third party doesn't stop such faulty operation or keep operating such operation, there is a major hazard to cause death or serious injury.
 Warning	It means if the operator or a third party doesn't stop such faulty operation or keep operating such operation, there is a potential hazard to cause death or serious injury.
 Attention	It means if the operator or a third party doesn't stop such faulty operation or keep operating such operation, there is a hazard to cause moderate or mild injury.

2)Warning sign and indication sign

Warning sign			
	Hands or clothes would be enmeshed.		Get injured when touch the running part.
	High voltage Electric shocking risk		High temperature Scalding risk
	Get injured when touch the driving part.		Put heavy object or hand on would cause damage.
Indication sign			
	Earth wire		The right direction as arrows

Safety precautions

The basic precautions:



Attention

1. Do read this manual and the instruction or other explanatory materials for this machine or the function before operating the machine. And keep the operation instruction in good condition in case you need any reference.
2. Not only the specifications for the machine but also other content is included in this article.
3. It is highly advised only the one who knows well the operations of this machine can operate this machine.
4. The daily repair and maintenance for this machine (not including the basic maintenance repair or other specific matters described in the manual) should be commissioned by our company or agency.
5. For general maintenance and repair (the specific matters in the manual) should be undertaken by the full-time service technician who should have been trained by our company. The parts used for maintenance (the specific matters in the manual) should be from our company.

Protector and warning sign

1. Make sure the protector is well installed in the right position before you operate the machine.
2. When remove the protector, make sure it is placed in the original position and the machine is in normal operation.
3. Make sure the warning signs on the machine is pasted correctly and eye-catchingly. If any of them is come off or stained, please contact our company for replacement.
4. Cannot operate the machine without protector. If you do that, it would cause accident.

Application

1. Do NOT us this machine for other purpose. Our company would not take any responsibility for the accidents thus caused.
2. Do NOT reform this machine. Our company would not take any responsibility for the accidents thus caused.

Training

1. To prevent accidents caused by unskilled operation, it is required that only the operators who are received specific operation training from our company and have appropriate knowledge can operate this machine.



Danger

If you have to open electrical equipment box, in order not to have electric shock accident, please cut off the power. For caution's sake, waiting for 2 minutes then open the box.



Attention

Power cut: It is referred to cut off the power switch and unplug the attaching plug or remove the power cable from the supply side. Similarly hereinafter.

Precautions when cut off the power:

1. When there is an abnormality, breakdown or power cut, please cut off the power.
2. In order to prevent accidents happening by starting the machine by accident, when overhauling, repairing and cleaning, make sure to cut off power before doing these.
3. When unplug the attaching plug, please hold the plug then unplug, not the cable.

Precautions in all aspects:

Transportation:

1. Make sure to take necessary security countermeasures to prevent the machine from invert or dropping when it is lifted or transported.

Unsealing:

1. Please read the all instruction on the transportation container
2. It is forbidden to cut the packing string with a knife.
3. Please take care of the shipping equipment.

Setting up:

1. In order not to cause any accident when moving the machine, please use a height adjusting device to hang the rolling wheel.
2. Please set up the machine at a horizontal place.
3. To prevent electric shock, electricity leakage and fire, the cable should be as accessory and connect to specified place.
4. To prevent electric shock, electricity leakage and fire, do NOT pull the cable with excessive coerce during running.
5. Set the attaching plug at the fixed place, when unplug it, hold the connector then unplug.

Before starting the machine

1. To prevent any human injury, before plugging in, make sure there is no damage, drop and loose for the connector or cable.
2. To prevent any human injury, do NOT put hands into the running part.
3. Make sure there is no any tool or other articles on the platform, or when the platform is moving, not only it would damage the articles; also it would deform the platform and cause failure.



Attention

Lubricating:

1. Use lithium base grease as lubricating grease
2. To prevent from inflammation and hypersensitivity, when the lubricating grease splashing into eyes or onto body, wash it immediately.
3. To avoid diarrhea and vomiting, when accidental ingestion, please go to see a doctor immediately.

Maintenance:

1. To prevent accidents from unskilled operation, repairing and debugging job should be undertaken by maintenance personnel who are familiar with the machine. Use our company part when replacing. Our company is not responsible for any accident by using parts not from us.
2. To prevent accidents or electric shock from unskilled operation, please appoint some personnel who have electrical expertise or the Technical staff from our company or distributor to repair or maintain the machine (including the wiring).
3. To avoid the accidents caused by pneumatic accident, before maintaining, make sure to remove the air source tube, let out the remaining air.
4. To avoid human injury, after repairing, debugging or replacing some parts, make sure the screws, nut don't loose.

Work environment:

1. To avoid the accidents by faulty operation, do NOT use in an environment affected by noise sources such as high-frequency welding machine (electromagnetic waves).
2. To avoid the accidents by faulty operation, do NOT use the machine when the power supply voltage is more than $\pm 10\%$.
3. To avoid the accidents by faulty operation, Please use the supply pressure of 0.6-- 1.0 Mpa.
4. For safe use, please operate the machine under the circumstances as below:

Operating environment temperature : $+10^{\circ}\text{C} \sim +35^{\circ}\text{C}$

Operating relative humidity 50% below (35°C)

90% below (20°C)

5. To avoid the accidents because of electric components breakage, when move the machine from high temperature place to warm place suddenly, there would be condensation. Please sweep away the water drop before turning on the power supply.
6. To avoid the accidents because of electric components breakage, unplug the attaching plug when it is thundering.

Precautions for safe use the HCT series SMT machine



Danger

1. To avoid the accidents by electric shock, do NOT open source the electrical equipment box when the power supply is on.
2. To avoid electric shock, do NOT run the machine when the earth wire is removed.



Danger

1. To prevent from human injury, do NOT run the machine when the protector is removed.
2. To prevent from human injury, please be careful, in order to avoid hair or clothes being enmeshed into the conveyor belt. In addition, do NOT wear glove.
3. To prevent from human injury, when maintaining (lubricating, debugging or daily maintenance), cut off the power supply.
4. To prevent from human injury, please use a residual-current circuit breaker on the power cable.请在电源线上使用漏电断路器。



Attention

The operation system for this machine is Windows XP. If there is some other software installed which is not used because thus machine, our company would not guarantee for its normal operation. Similarly, if transfer ,rename or copy the document in the hard disc drive, we cannot guarantee its normal operation. Sorry for the inconvenience.

- Just in case it happens, it is required to replace a new HDD, which would lead data loss, therefore, this point should be paid special attention.

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Chapter One Equipment Overview

1-1 Preface

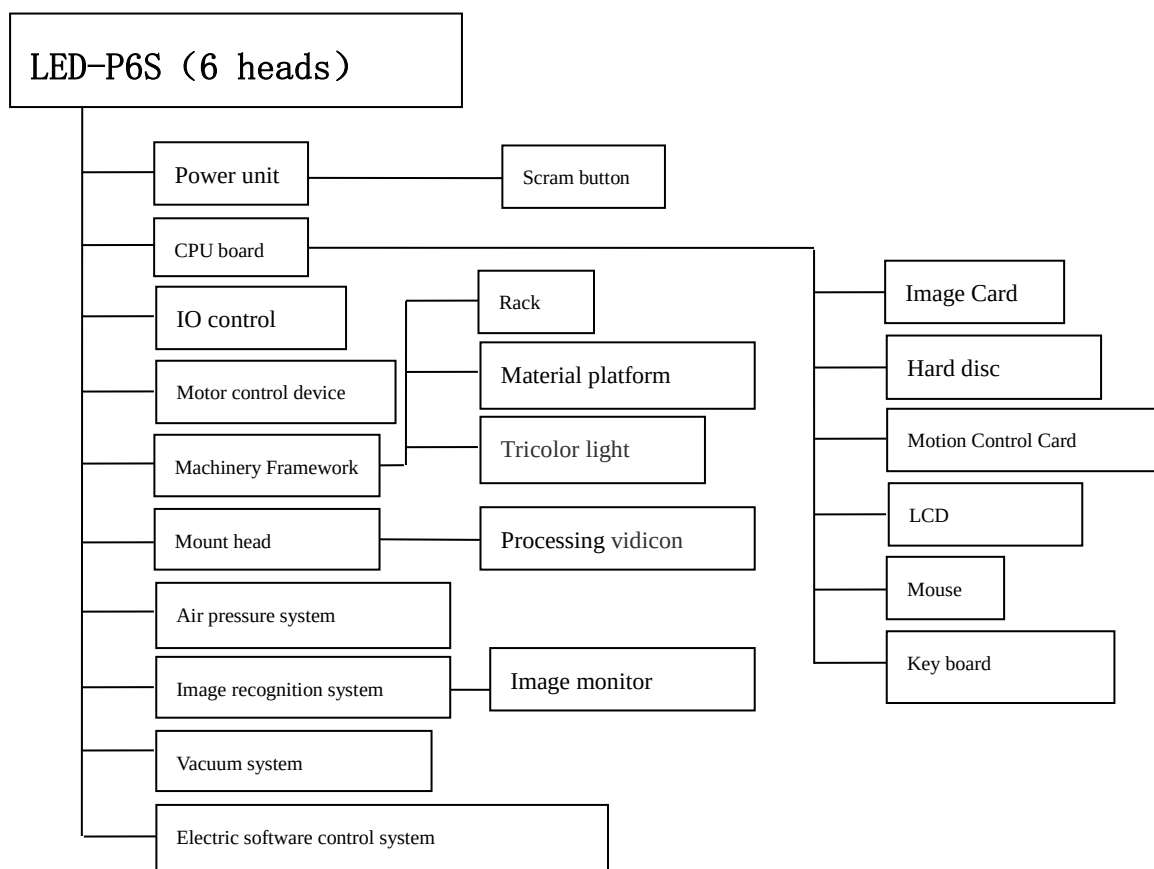
“SIGPlace LED-P4S/SIGPlace LED-P6S” (hereafter referred to as ‘SIGPlace Chip Mounter series’) is a well-designed equipment for LED mounting. Normal configuration with image vision alignment SIGPlace LED-P4S (four heads), high-level configuration with image vision alignment SIGPlace LED-P6S (six heads), compatibility, reliability, maintainability and safety. SIGPlace Chip Mounter series is using the Windows XP as its operation system software (basic software), easier to operate, mainly applied to mount LED light, can also be used to mount slim shape chip component and some QFP,CSP and etc.

Below is the detail for SIGPLace LED-P6S:

1-2 Feature

- 1): Use Windows XP, easy to operate
- 2): Nozzle XY using high precision ball screw to support, less abrasion and noisy, higher accuracy, longer life
- 3): Use processing vidicon function, easier, more convenient and faster, increase the production efficiency.
- 4): With the component position angular deflection and auto correction identifiable function, ensure the paster accuracy.
- 5): With negative pressure detector, can prompt the null in production process.

1-3 System constitution



1-4 Technical parameters

1):Max mounting circuit board size :	1200mm (L) × 310mm(W)
2):Max component height:	6mm
3):Production capacity:	20000 CPH (Not including the manual feeding and blanking time)
4):Component Centering method:	Vision system+ processing vidicon
5):Mounting accuracy:	±0.1mm
6):Component scope :	0603~7474, some special component IC/QFP/CSP
7):Dimensions:	2000(L)×970(W)×1350(H) mm
8):Power supply:	220V, 50Hz
9): Air supply Pressure:	0.55Mpa
10):Feeder(s) :	14 pcs (Maximum)
11):Power:	2000W
12):Placement head:	6 heads
13): Rotor angle:	Self-synchronizing rotor angle
14):X/Y run mode:	Linear linkage
15):Operation system:	WINDOWS XP
16):Machinery accuracy:	0.03mm
17):Component shortage warning:	KITA negative pressure detective system
18):Weight:	approx 850KG

1-5 Equipment overview

1) Dimensions as FIGURE 1、FIGURE 2

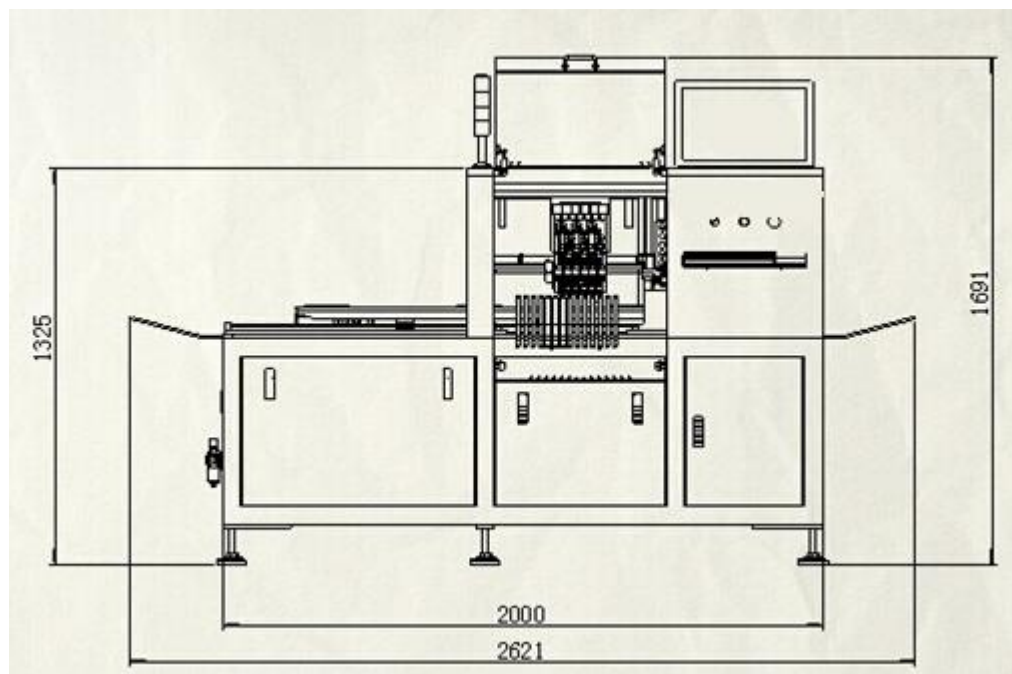


FIGURE 1: Front view

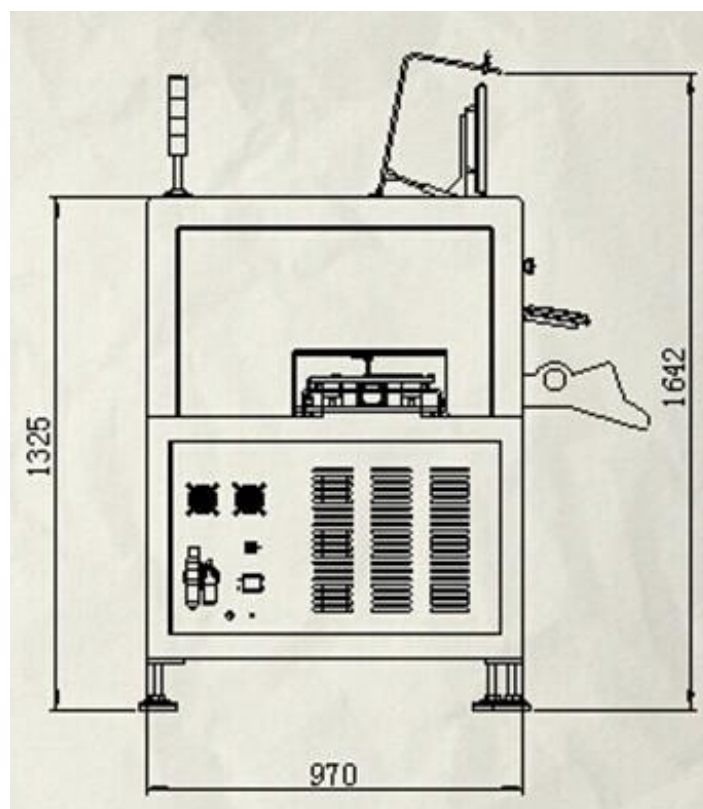


FIGURE 2: Side view

2) Equipment composition

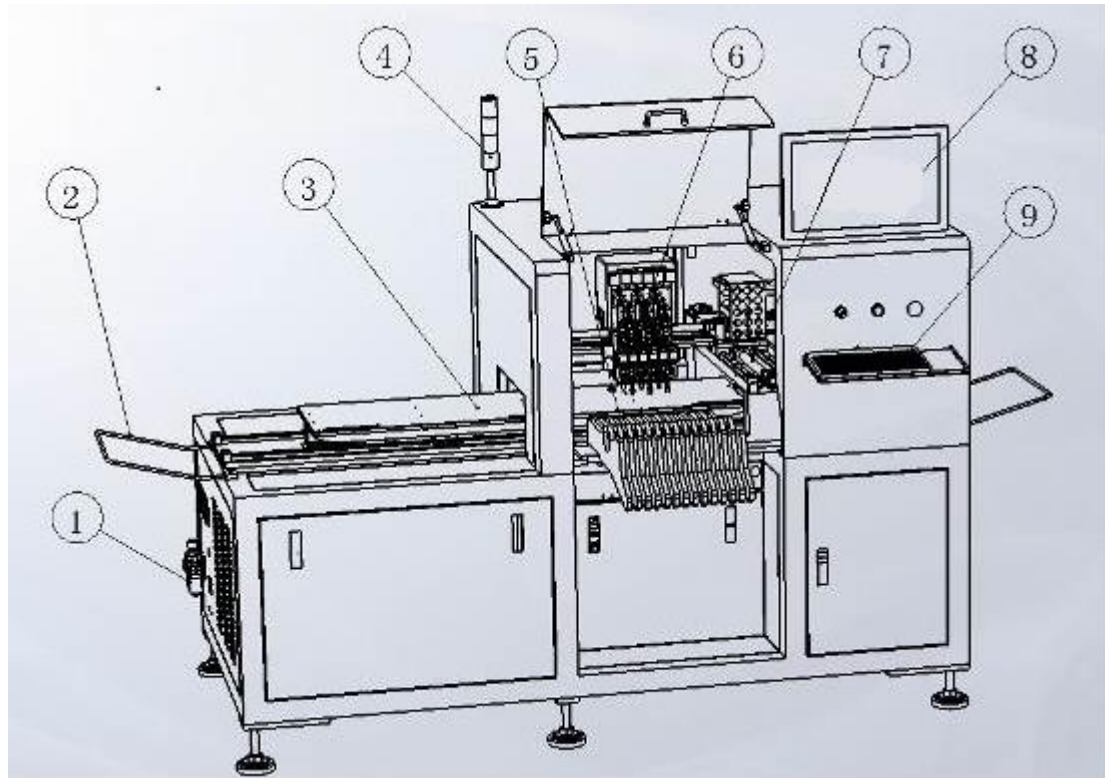


FIGURE 3

- | | | |
|-------------------------|---------------------|---------------------|
| 1 Pneumatic filter | 2 Protective device | 3 Platform |
| 4 Signal light | 5 Feeder | 6 Nozzle components |
| 7 XY suction components | 8 Displayer | 9 Key board |

3) Emptying platform brief introduction

In order to have a high mounting accuracy, the platform is to adopt servomotor to drive the high precision ball screw, with theoretical accuracy 0.002mm and real accuracy 0.01mm, also make the whole platform planeness is within 0.5mm.

Its specific construction is as FIGURE 4.

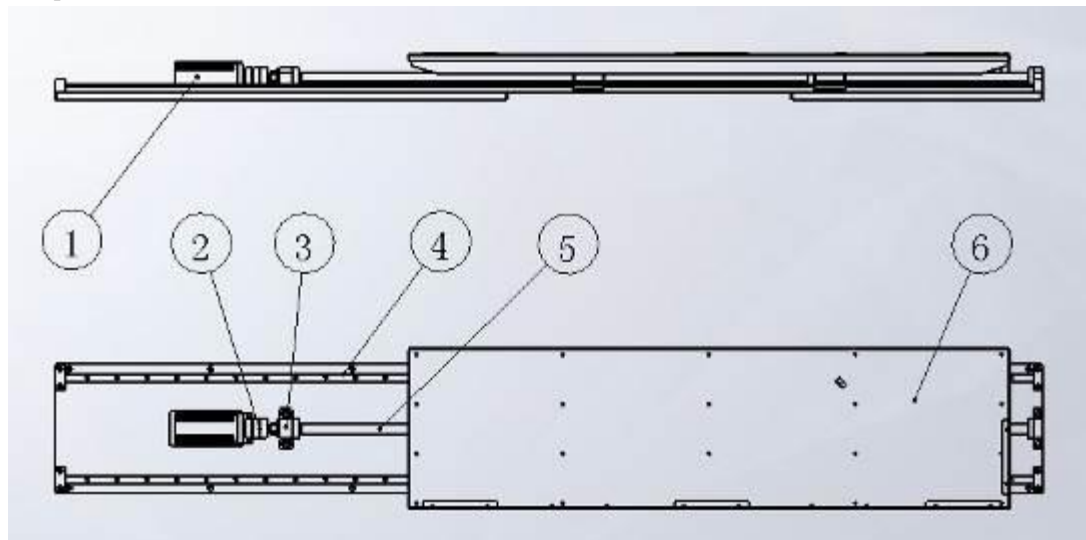


FIGURE 4: Emptying platform diagrammatic drawing

- | | | |
|----------------|---------------------------|--------------------|
| 1 Servomotor | 2 Shaft coupling | 3 Ball screw block |
| 4 Linear guide | 5 Ball bearing lead screw | 6 Platform |

4) Nozzle XY axis brief introduction

X&Y axis are using the servomotor to drive the high precision ball screw, its theoretical accuracy is up to 0.002mm, and with a linear guide rail supporting, guarantee higher accuracy, less noisy and less attrition, and longer service life. Its general construction is as FIGURE 5.

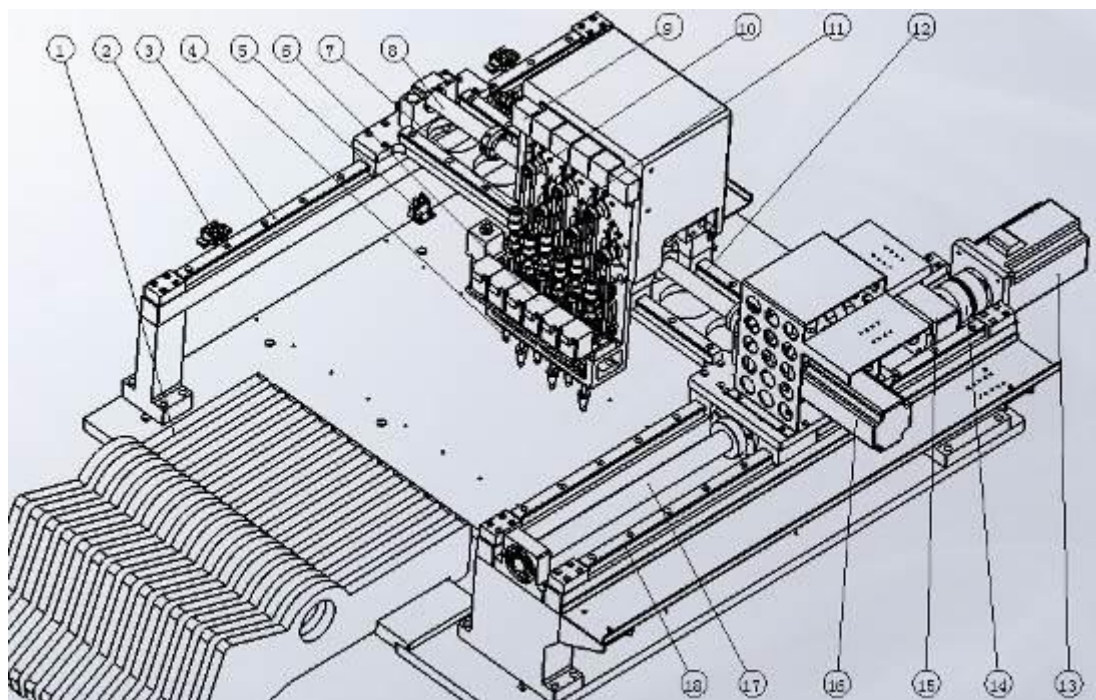


FIGURE 5: XY shafts construction diagrammatic drawing

- | | | |
|-----------------------------|------------------------------------|-------------------------------------|
| 1 Feeder | 2 Y shaft limitation | 3 Y shaft linear guide |
| 4 Nozzle | 5 Platform base point | 6 Rotary motor |
| 7 Rotary joint | 8 X shaft Ball bearing lead screw | 9 Negative pressure detector switch |
| 10 Z shaft synchronous belt | 11 Z shaft base point | 12 X shaft linear guide |
| 13 Y shaft servomotor | 14 Shaft coupling | 15 Ball screw block |
| 16 X shaft servomotor | 17 Y shaft ball bearing lead screw | 18 Y shaft linear guider |

5) Nozzle Z axis and angle T axis brief introduction

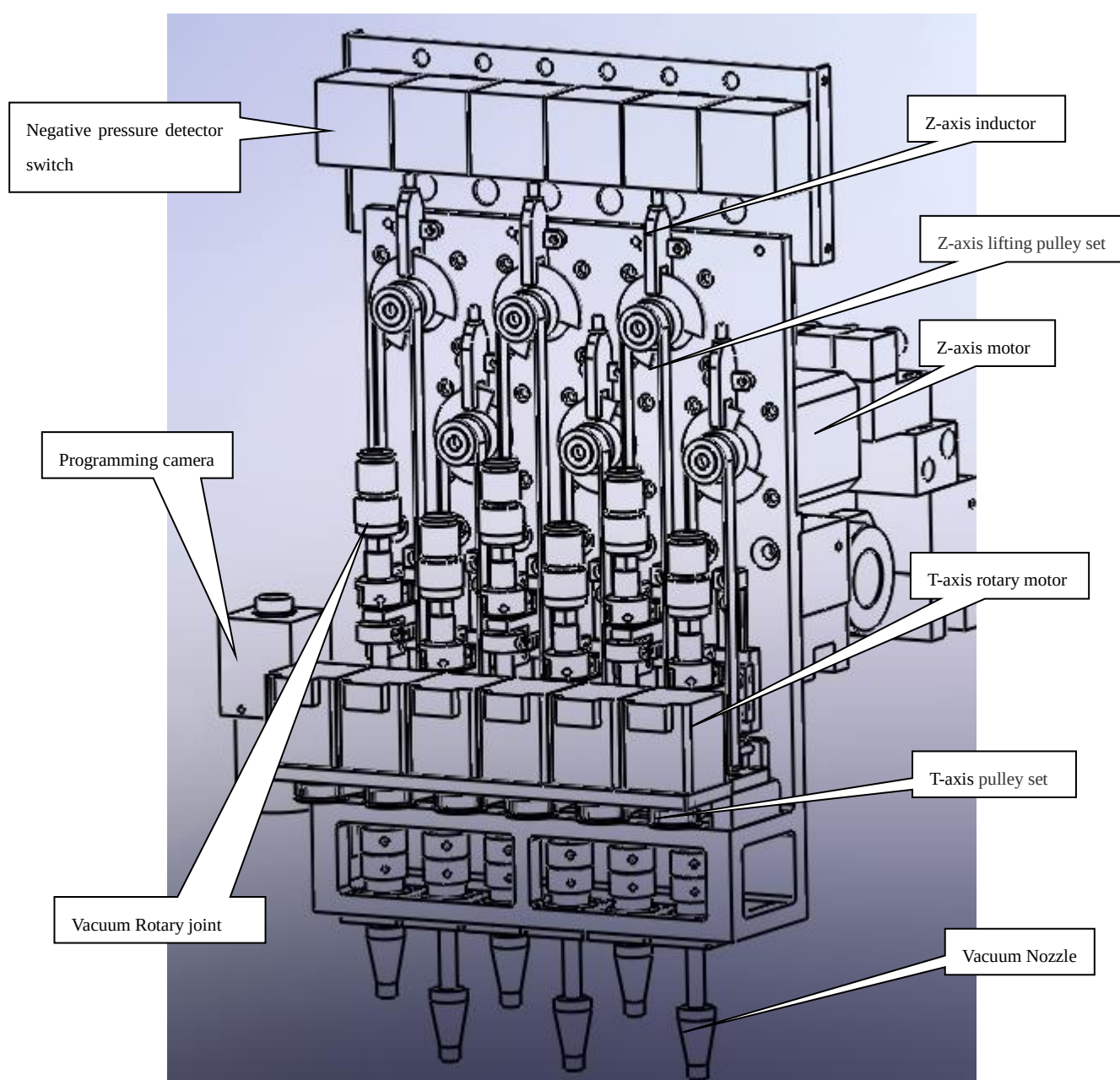


FIGURE 6: Z- Axis construction diagrammatic drawing

The up and down movement of the vacuum nozzle is driving by the Z shaft motor pulley, when the motor drives

the pulley counterclockwise rotating, the nozzles go down, when the motor drives the pulley clockwise rotating , the nozzles go up. The nozzles are using a linear guide way which can guarantee the up and down accuracy and the stability for mounting.

The suction rod is hollow, vacuum can go through it to the nozzle to suck the material.

1-6 Air-operated system overview

Our mounter don't provide air source, the air source would be prepared by the buyer's own. The minimum pressure of the air source should be 0.55MP—0.7MP. One would go through slide switch (a switch of the machine air source), water separation filter (filtering the fluid oil in the air, condensate water and impurity, filtering accuracy is 20 μ m), and solenoid valve(controlling on-off), to the cylinder of feeder, another one would go through the vacuum generator to generate vacuum, the vacuum would go through a rotary joint the suction nozzles.

See Annex 2 (air circuit schematic)

1-7 Vacuum system overview

The mounter is provided air by air source. Passing through solenoid valve(controlling on-off of vacuum), going into the vacuum generator and make the vacuum happen. The degree of vacuum could be as high as 80 KP. It would come through a high-speed rotary joint and a hollow rotary motor to the suction nozzle. There is a vacuum pressure switch, when there is no material or wrong position for the suction nozzle, the vacuum degree would be lower that set value. The switch would send signal to the controller, and set a warning.

See Annex 2 (air circuit schematic)

Chapter Two Brief Introduction & Installation of Operation System

2-1 Brief introduction of the software function

1) This is a bundled software for the LED-P6S automatic LED mounter Its operation interface is English/Chinese, very easy to use, suitable for majority of LED SMT factory.

2) Software operating environment

Hardware environment:

CPU: Pentium 2.4 or above

RAM: 512M or above

Hard disk: 5.6G at least

Power: 350W

Software environment:

Windows XP

2-2 Brief introduction of installation interface

1) Driver installation:

Copy the HC-setup file which is provided by us to Disc E, as FIGURE 1

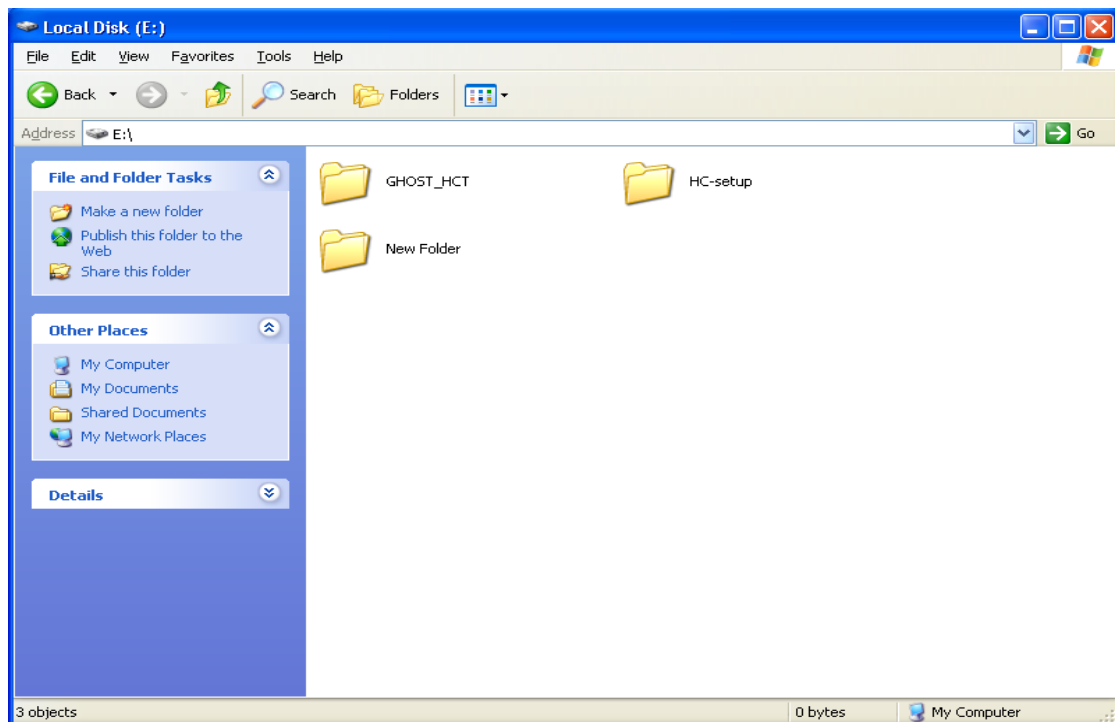


FIGURE 1

Extract and open HC-setup folder, as shown in FIGURE 2 four files and install the driver according to directions, can be installed directly.

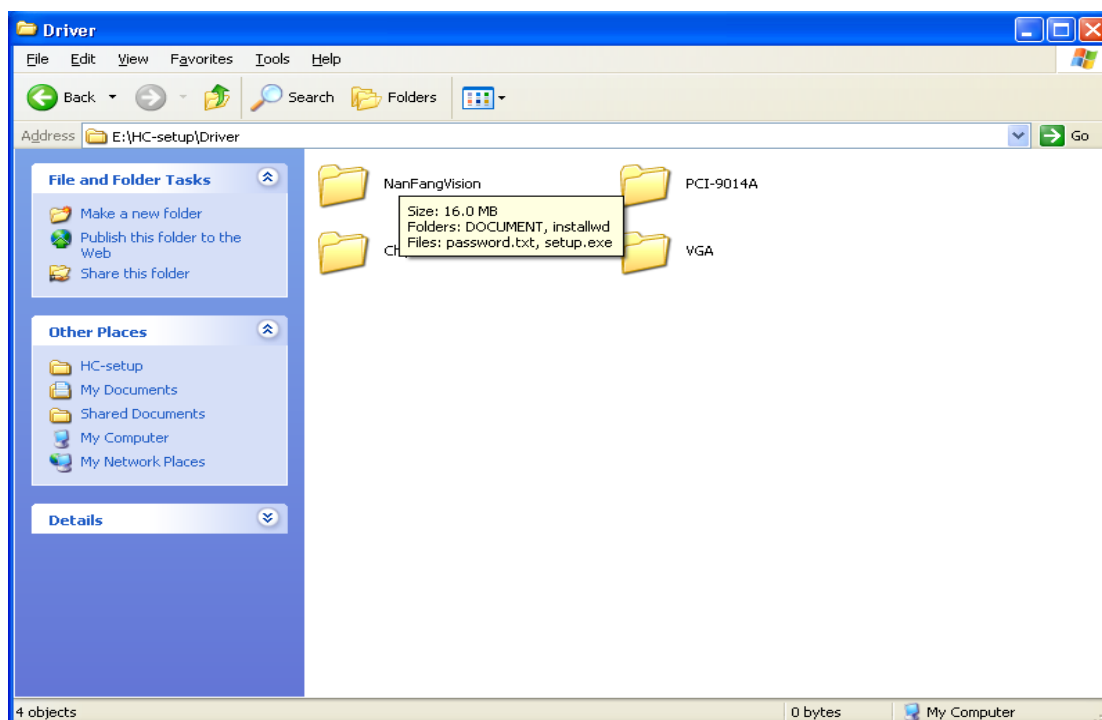


FIGURE 2

REMARK: Image Driver Installation password is [sjx20121225](#)

After a successful driver installation, the device manager would show as FIGURE 3:

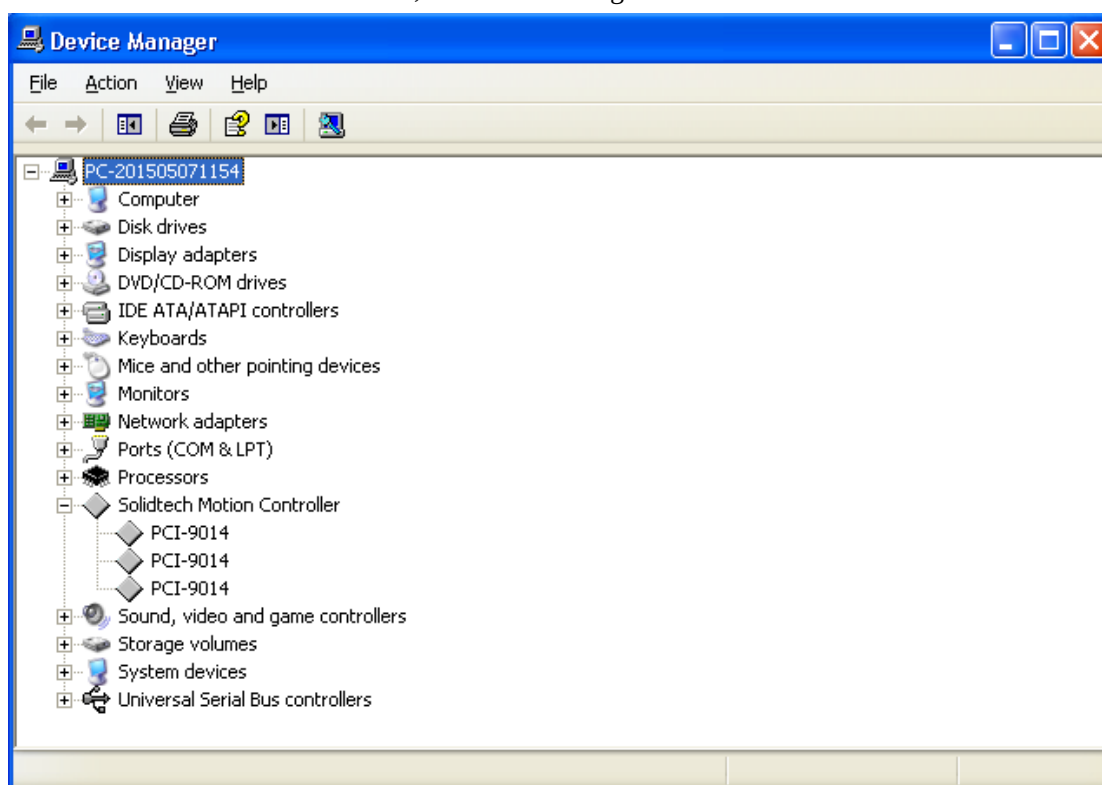


FIGURE 3

2) Operation software installation

This operation software is green software, no need to install. Copy the HC03 folder in the HC-setup folder to D:\ root directory, as FIGURE 4.

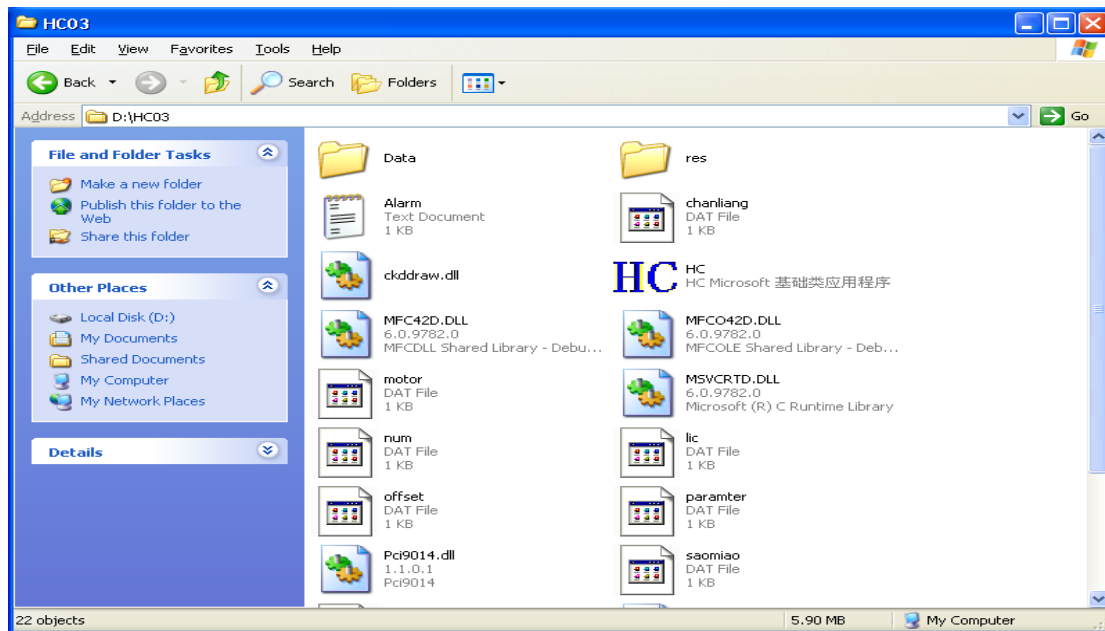


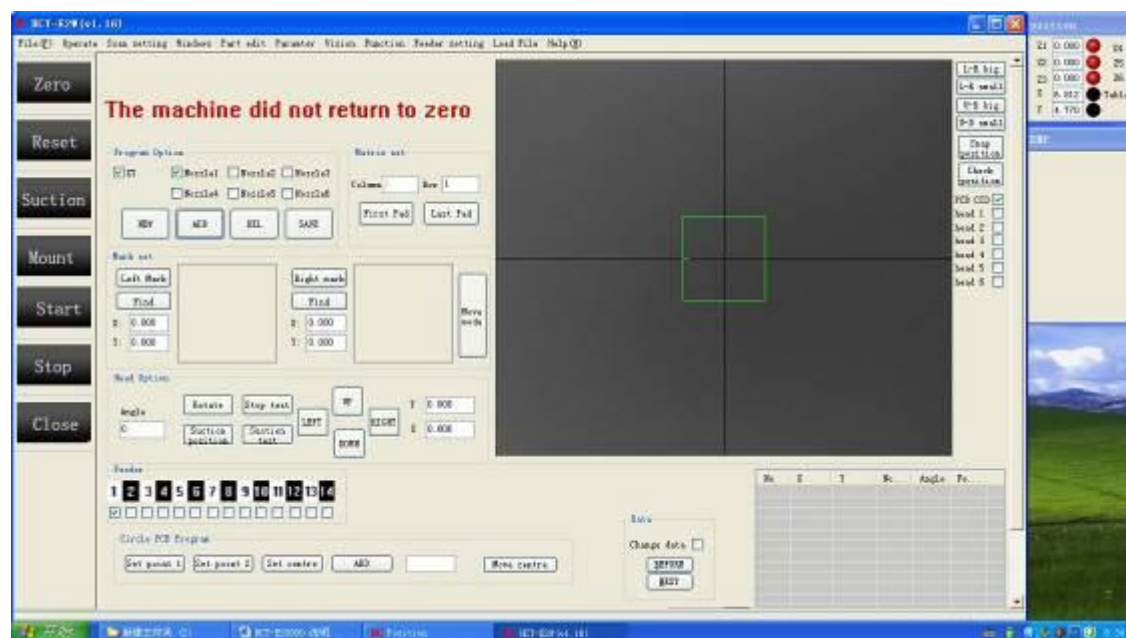
FIGURE 4

Open HC03 file, run HC program document then you can run the operation software.

Chapter Three Manufacturing & Production Procedure

3-1 Run debugging and Brief introduction of software interface

1) Software running main interface



① Left column Main Button introduction

Zero	Control motors find zero-point
Reset	Control motors back to zero-point
Suction	Adjust suction parameters & Pin set
Mount	Adjust mounting parameters
Start	Start production
Stop	Stop/Pause production
Close	Exit operation software

② Options bar Main Option introduction

➤ File

- 1) **“Open file”**: TO open the edited & saved production files
- 2) **“Open another file”**: TO merge files

Note: When one PCB cannot program exactly at once or too cumbersome to program at one time, can separately program and named file as “No. 1”, “No. 2”, ...and so on. After program, “Open file” to open the “No.1” file, and then “Open another file” to open “No.2” file, (or “No.3” file...). Then click “Save”, “No. 1” and “No. 2” files will be merged into one file and saved in the “No. 1” file, then you can get a whole PCB production file now. When start production, only need to open the “No. 1” file directly, do not need to merge again.

➤ Operate

- 1) **“I/O test”**: TO test and adjust parameters
 - ① I/O **“In”**: When the corresponding sensors are sensing, the corresponding I/O “In” flashing red, can be used to test whether the corresponding sensor is damaged.
 - ② I/O **“Out”**: When checking the corresponding I/O “Out”, the corresponding components will move, can be used to test whether the corresponding components damage and adjust parameters.

➤ Windows

- “Vision frame”**: To just the positioning frame size
“Motor position”: To bring up the dialog
“Show pad”: To show programmed pad

➤ Part edit

- 1) Why to make a Component Pattern?

When using the **“Vision check”** for production, the nozzle will go through the camera to take picture (for alignment) after suction the components, the picture will automatic compare with the built Component Pattern. When discover angular deviation with built Pattern, the nozzle will rotate for correction. (Built Component Pattern must be exactly right, otherwise the board will be mounted with deviations.)

- 2) How to build a Component Pattern?

Take No. 1 nozzle as example.

- a. Move No.1 nozzle to the feeder position to pick up the component, click **“Snap position”** (on the right side of computer screen), the nozzle will move to the camera position, you can see the component pattern on the image display area.
- b. Regulate the light, to observe whether there is an angular deviation. If there is deviation, need to rotate the angle, and then to build the original pattern.
- c. Turn to the **“Pattern mode”**, click and drag with the right mouse button to frame the component image in the image display area, note the edge of the drawn frame box should space about 2mm from the component pattern, and the center of the drawn frame should match the component image center, click on the **“Set pattern”** in the drop-down menu, observe whether displayed templates is appropriate.
- d. If inappropriate, to change Pattern gray level in the Vision parameter, such as pattern contour too much increase the parameter, too little decrease the parameter.
- e. After completed pattern setting, click the drop-down menu to **“Save Pattern”**, name and set the

thickness. Finally to choose this pattern from the “Vision” “Select part” in corresponding Feeder selection.

PAY ATTENTION

Drag the right mouse button on the displayed image to frame, must be from the top left corner on the bottom right corner

Pay attention to the component pattern according to the outline, too much/ too little is not reasonable

➤ Parameter

“**Parameter setting**”: To bring up “**Suction setting**”

“**High parameter**”: To adjust the equipment running speed. Other parameters should be adjusted under the guidance of our company technical engineer. (Password: three space key and z)

➤ Vision: To choose the Component Pattern & adjust camera parameters

➤ Function

“**Nozzle_Feeder**”: To change or distribute Nozzle and Feeder

“**Optimize (same part)**”: Using in the case of a single component for production, allow to mount in the shortest distance

➤ Feeder setting: Use programming camera to find the the first component in the feeder and “**Set feeder position**”

Note: Because of the route limitation, the camera can max move to No.10 feeder, you can set No.1-10 feeder position here only.

➤ Load file: To import CSV file of PCB, need some modify before production.

③ Operation software main interface segmentation introduction

FIGURE 1 Programming options for normal PCB

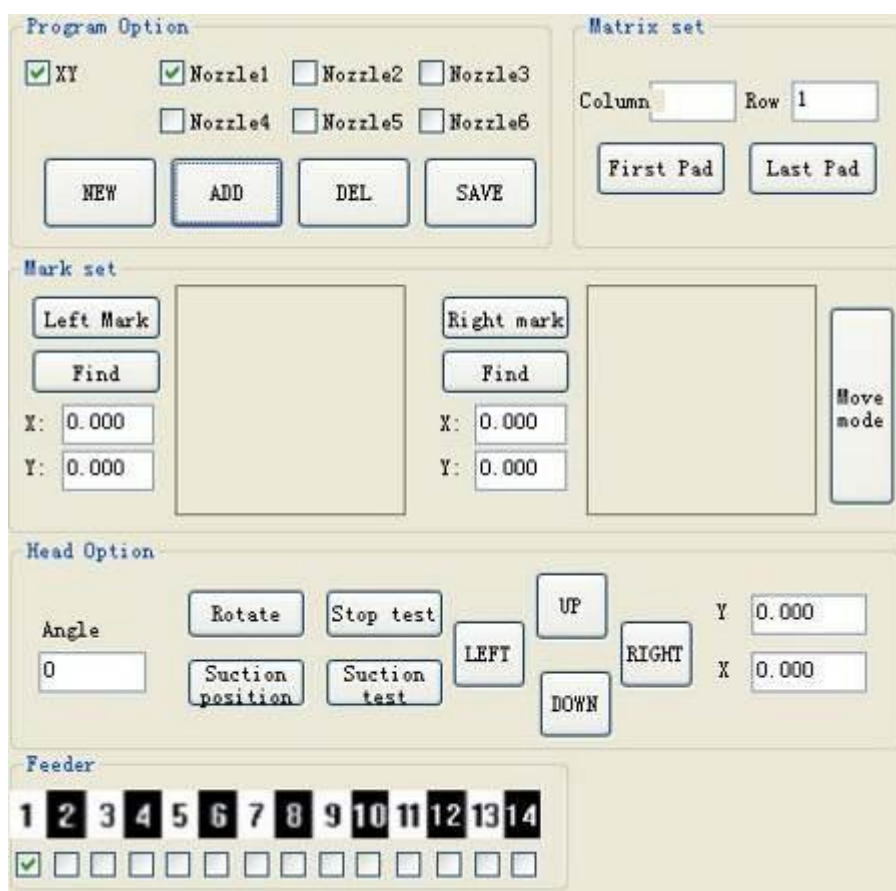


FIGURE 1

FIGURE 2 Programming options for Round Board (Circle PCB Program)



FIGURE 2

Note: Only for the points on the round PCB is uniformly distributed and symmetrical.

Programming:

- Find two symmetrical points about the center of the round board, locate one of them and click “**Set point 1**”. Move the suction nozzle (keep platform not move) to locate another symmetrical point and “**Set point 2**”. And then click “**Sets center**”.
- After setting, click “**Move center**”, the camera will automatically find the center of the board. Count the points of the first lap and fill the number to the blank box. Locate any one point of the first lap, click “**Add**”, there will be a pop-up dialog “**Auto angle set**”, click Yes, the points location of first lap will add to the list.
- And then click “**Move center**” to go back to the center, follow the previous step to add other laps. NOTE: After added each lap, the camera need to go back to the center firstly and then go to add another lap.
- Finally to set Mark point and save to finish the round board programming. Shown as FIGURE 3.

No.	X	Y	No...	Angle	Fe...
1	371...	244...	0	0	1
2	366...	254...	1	300	2
3	355...	254...	2	240	3
4	349...	244...	3	180	4
5	355...	234...	4	120	5
6	366...	235...	5	60	6
7	383...	244...	0	0	1
8	382...	253...	1	336	2
9	376...	261...	2	312	3
10	368...	266...	3	288	4
11	358...	267...	4	264	5
12	349...	264...	5	240	6
13	342...	258...	0	216	1
14	338...	249...	1	192	2
15	338...	240...	2	168	3
16	341...	231...	3	144	4
17	348...	224...	4	120	5
18	357	221	5	96	6

FIGURE 3

FIGURE 4 Motors position display & Motors corresponding origin sensor signal

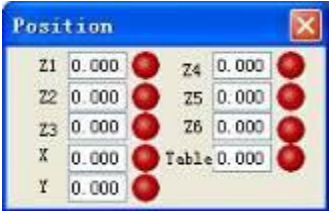


FIGURE 4

FIGURE 5 Alignment camera options



FIGURE 5

FIGURE 6 Production calculation area

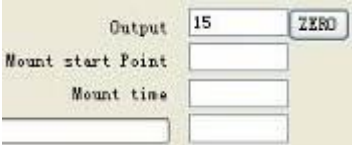


FIGURE 6

FIGURE 7 “Move mode” & “ Pattern mode” toggle button

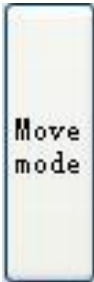


FIGURE 7

FIGURE 8 Button to Adjust the positioning frame size

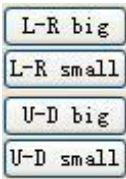


FIGURE 8

FIGURE 9-11 Buttons to adjust nozzle/angle/feeder if make any mistakes during programming



FIGURE 9



FIGURE 10

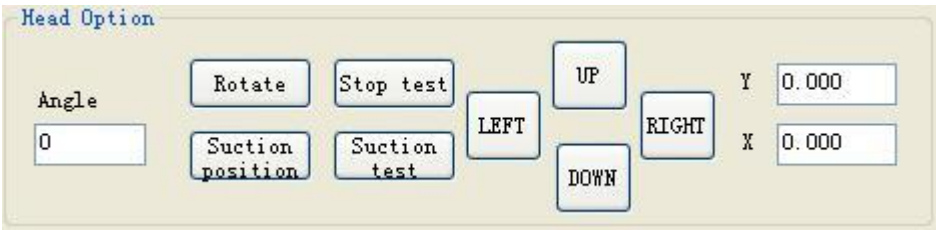


FIGURE 11

FIGURE 12 Image display area & Camera switching operation options

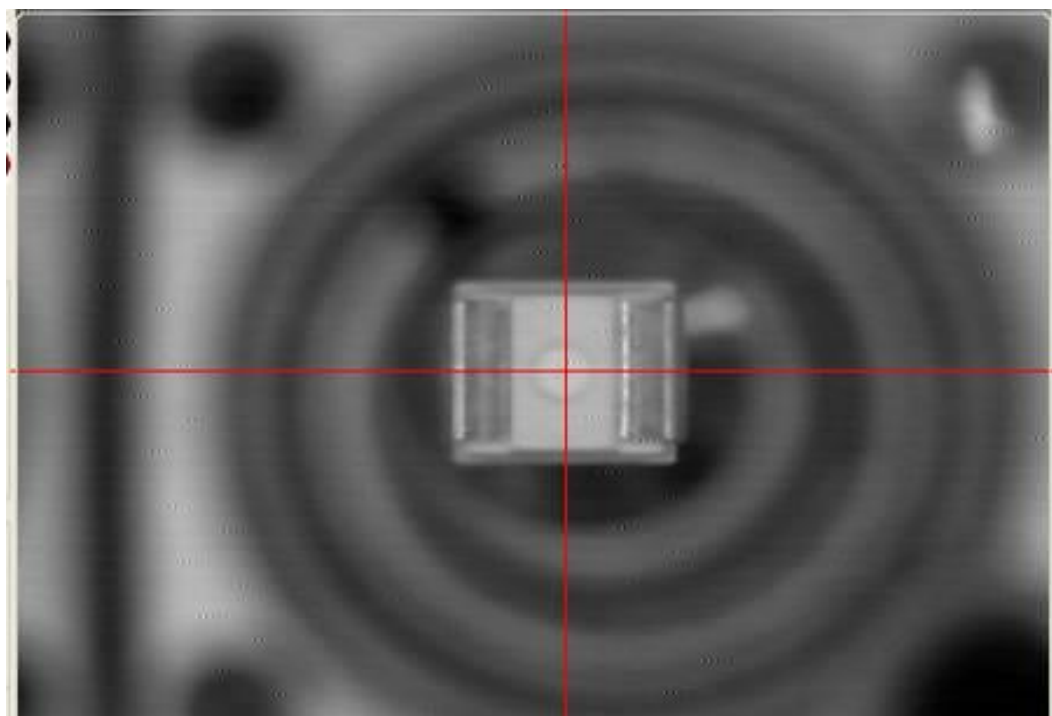


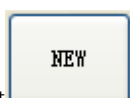
FIGURE 12

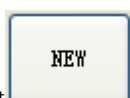
3-2 Software operation

1) Shortcut key

- 1) Keyboard '<' and '>' is to move the platform left and right
- 2) Keyboard '←' and '→' is to move the nozzle components in X axis direction
- 3) Keyboard '↑' and '↓' is to move the nozzle components in Y axis direction
- 4) Right mouse click on the camera image position, it would move this position to the image center.
- 5) '-' '+' key can change the speed of Shortcut moving.

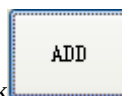
2) Programming operating:



- ① Firstly, select  to clear the file in the list area and build a new file.

Add point location: Move the platform or nozzle components to locate **the first point** of the PCB in the center of the image, adjust the positioning frame size to match. If there is an angel, just input the angel degree. Then select the suction nozzle and the corresponding feeding station (feeder). As below screen-shot.





After selecting the suction nozzle and feeder, click **ADD** to add this point location into the file list.

Shown as below:

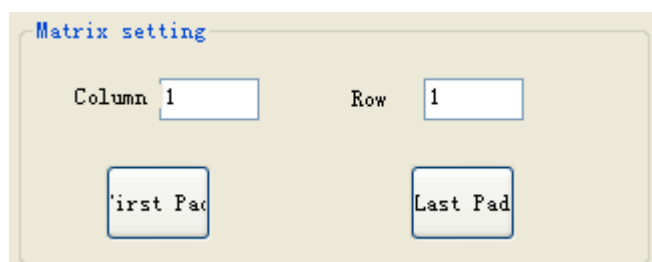
NO.	X	Y	No...	Angle	Fe...
1	160...	557...	0	0	1
2	160...	557...	1	0	3

② Finish the setting of first point, the suction nozzle and feeder would automatically shift to next ones.

Repeat the operation as previous, move the suction nozzle to locate the second point of the PCB, add the data into the file list.

③ The points on the PCB are always regular, choose equidistant unit as matrix permutation.

Firstly, it is required to add all points of the first unit, and then proceed a matrix setting as below:



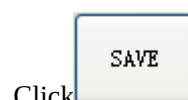
Input the amount of equidistant unit into Matrix setting.

① Set **Row** firstly (Column sets 1, Row sets the amount of units), locate one point in the first unit, set it as “**First Pad**”. And then locate the corresponding point in the last unit of matrix, set it as “**Last Pad**”.

② Then set **Column** (Row resets to 1, Column sets the amount of units), follow the setting step of Row.

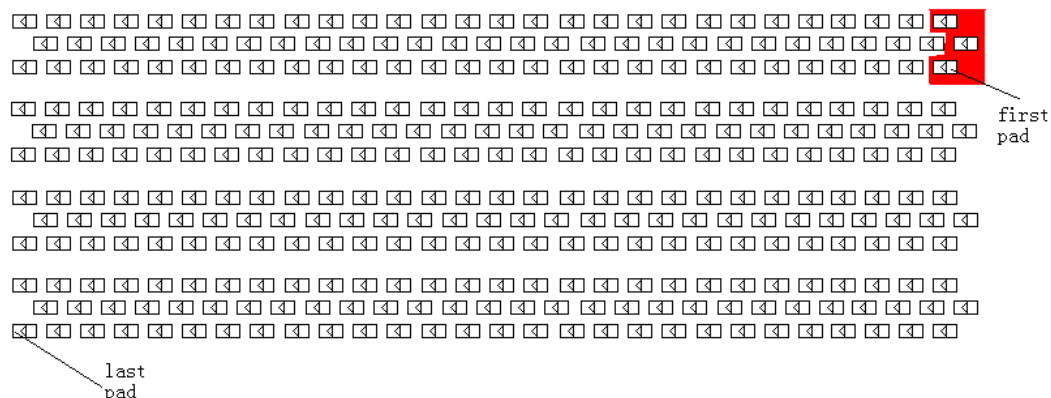
After setting, all the points location in the whole matrix permutation would be shown in the list. Shown as below:

NO.	X	Y	No...	Angle	Fe...
1	160...	557...	0	0	1
2	160...	557...	1	0	3
3	161...	557...	2	0	5
4	161...	557...	3	0	7
5	161...	557...	0	0	1
6	161...	557...	1	0	3
7	161...	557...	2	0	5
8	161...	557...	3	0	7
9	161...	557...	0	0	1
10	161...	557...	1	0	3
11	161...	557...	2	0	5
12	161...	557...	3	0	7
13	161...	557...	0	0	1
14	161...	557...	1	0	3
15	161...	557...	2	0	5
16	161...	557...	3	0	7



Click **SAVE**, save all the data.

Example:



This is a four pin board, there are 84 LED chips each single board. It can be programmed as a unit for the 3 LED chips (like in the red area), add first 3 points location from top to bottom. Then take the undermost point as the “**First Pad**” of the matrix setting. 84 LED chips each single board would be 28 units in Column and 4 units in Row when setting. The setting for matrix is as below:

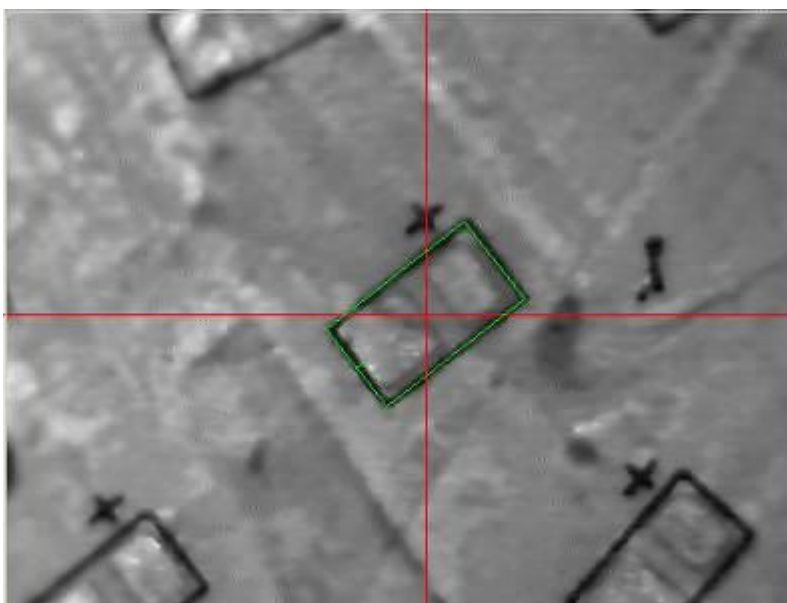
Matrix setting

Column Row

After setting the “First Pad” (the undermost point of the first unit), move the platform or suction nozzle to locate the corresponding point of the last unit and set “**Last Pad**”. Save the data after finish the setting. The program procedure is finished.

3) Programming for the irregular PCB

Because the LED chip or components on an irregular PCB are not regularly arranged, so it is required to add every point location.



As shown above, locate the point on the bonding pad, check the image display area to make sure the image on the center, adjust the positioning frame (green box) size to match the pad of component. Drag the green box with the mouse to have the minus sign of the box to match the negative direction of the pad, select the suction nozzle and feeder and then add this point location.

4) Mark point setting

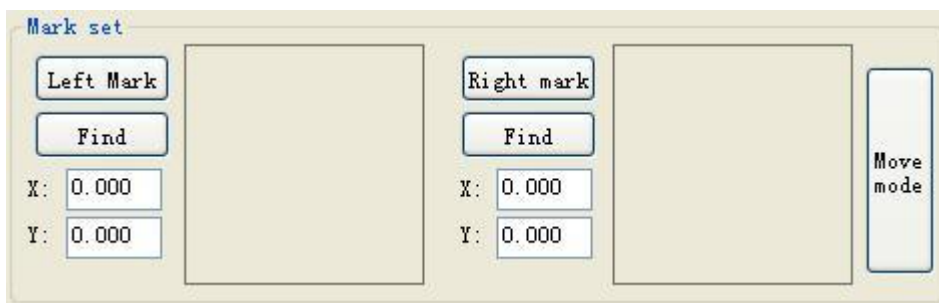
① Why need Mark point?

Mark point is for fine-alignment position, it plays a decisive role for whether a PCB can be mounted exactly or not. The camera firstly identify two Mark points when starting production, and then calculate and recognize the components location for mounting. If the Mark point alignment failed or error, the corresponding components position will be wrong.

② How to set Mark point?

DO NOT MOVE THE PCB and go to “**Mark set**” after added all points location of the PCB.

- Observe the PCB to find two symmetrical points (**MUST top left & bottom right**) as Mark point setting
- Move the camera to locate the top left point, make sure the point image is shown in the center. Change “**Move mode**” to “**Pattern mode**”, click and drag with the right mouse button to frame the Mark point image in the image display area, then click “**Left Mark**” to set. After setting, click “**Find**” to check whether there is a green cross symbol in the center of the Mark point image, if yes, save the Mark point; If not, need to reset.
- The bottom “**Right mark**” point setting is same as the left one.



NOTE: If too much gray scale (shown as green) on the Mark image, need to set high Mark gray level in the Vision parameter.

5) Parameter setting

This software can control 6 pcs of suction nozzle and can program & operate 14 pcs of feeder, so it is required to set the pick position and suction height for each feeder.



Click here to start suction setting (FIGURE 1)

	Suction height (mm)	Delay (ms)	Suction X (mm)	Suction Y (mm)
Feeder1	13.5	10	254.92	421.1
Feeder2	13.5	10	229.92	421.1
Feeder3	13.5	10	204.92	421.1
Feeder4	13.5	10	179.92	421.1
Feeder5	13.5	10	154.92	421.1
Feeder6	13.5	10	129.92	421.1
Feeder7	13.5	10	104.92	421.1
Feeder8	13.5	10	79.92	421.1
Feeder9	13.5	10	54.92	421.1
Feeder10	13.5	10	29.92	421.1
Feeder11	13.5	10	4.9199	421.1
Feeder12	13.5	10	-20.08	421.1
Feeder13	13.5	10	-45.08	421.1
Feeder14	13.5	10	-70.08	421.1

Pin set

Pin number: 1
Pin distance: 0

move height: 3
Suction check: 0
Check delay: 10
Vision check: 0
Check times: 2
Suction at same: 1
Use mark: 0

Cancel OK

FIGURE 1

“Suction height”: The height of suction nozzle go down to suck the components from the feeder

Setting: Select ☒ Nozzle1 (as example), click “Suction position” to make the suction nozzle to move to the feeder position. Go to Operate drop-down menu, click “I/O test”, check the corresponding feeder to open the cover. Then press the ‘ ↓ ’ key till the suction nozzle just touch the LED chip surface, check the motor position of suction nozzle, input the position data with added 0.5mm to the Suction height. Follow the previous steps to set other feeders.

“Suction position (XY)”: The position coordinates of corresponding feeder

Setting: Select ☒ Nozzle1 (as example), click “Suction position” to make the suction nozzle to move to the feeder position. Go to **“Operate”** drop-down menu, click **“I/O test”**, check the corresponding feeder to open the cover. Then press the ‘ ↓ ’ key till the suction nozzle just touch the LED chip surface, observe whether the suction nozzle is sucking in the middle of the LED chip. If not, move the X axis or Y axis direction to adjust the parameter. Follow the previous steps to set other feeders.

“Move height”: The height of suction nozzle (Z axis direction) during sucking & moving, can help to improve production efficiency. **Note**: Please do not set this valve too high, or the nozzle rod may crash

“Suction check” To check whether sucking materials successfully, there is alarm if sucking failed,

help to prevent missing mounting (Set “1 “to ON, “0” to OFF)

“**Vision check**” To check Vision Alignment (Set “1 “to ON, “0” to OFF)



Click here to start mounting setting (FIGURE 2)

Mount height	
Feeder 1	9.7
Feeder 2	9.5
Feeder 3	9.5
Feeder 4	9.5
Feeder 5	9.5
Feeder 6	9.5
Feeder 7	9.5
Feeder 8	9.5
Feeder 9	9.5
Feeder10	9.5
Feeder11	9.5
Feeder12	9.5
Feeder13	9.5
Feeder14	9.5

mount delay	20
Release Vacuum delay	5
Mount wait height	2
Mount rise delay	10

OK

Cancel

FIGURE 2

“**Mount height**”: the height of suction nozzle go down to mount the components to the PCB according to different feeders

Setting: Move the suction nozzle with sucked LED chip to the point location of PCB, select ☒ Nozzle1 (as example), remove ✓ of ☒ XY, press the ‘ ↓ ’ key till the LED chip just touching the PCB, check the motor position of suction nozzle, input the position data with added 0.5mm to the Mount height. Follow the previous steps to set other feeders.

☒ XY	☒ Nozzle1	☐ Nozzle2	☐ Nozzle3
	☐ Nozzle4	☐ Nozzle5	☐ Nozzle6

NEW

ADD

DEL

SAVE

6) Production operation



Click  to enter production standby mode. Put the PCB well and then press the 'Start' button on the machine operation panel. (Put the negative direction of PCB to the left side)

① If setting the Mark point, the camera will automatically recognize the two Mark points, after recognition the machine will go into production.

② If not setting the Mark point, press the 'Start' button on the machine operation panel, go to First pad setting, move the suction nozzle to find the first point and set.

And then press the 'Start' button on the machine operation panel again, go to Final pad setting, move the suction nozzle to find the corresponding final point and set.

After setting, press the 'Start' button on the machine operation panel, enter into production mode.

Press 'Pause' button to pause mounting during production. You can also do fine adjustment of the SMD position; or adjust the suction nozzle according to the LED welding spots actual position.

7) If there is a suction nozzle or feeder fail to work, you can use the “Nozzle Feeder” application function, just input the corresponding nozzle or feeder number to match corresponding replace nozzle or feeder. **Please pay attention that No. 5, 6 nozzle cannot match No. 1, 2, 3, 4 feeder, otherwise will cause crash.**

Nozzle & Feeder application

Nozzle

1	2	3	4	5	6
1	2	3	4	5	6

Feeder

1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	2	3	4	5	6	7	8	9	10	11	12	13	14

NOTE: pay attention to Nozzle and Feeder

OK Cancel

Chapter Four FAQ for the machine

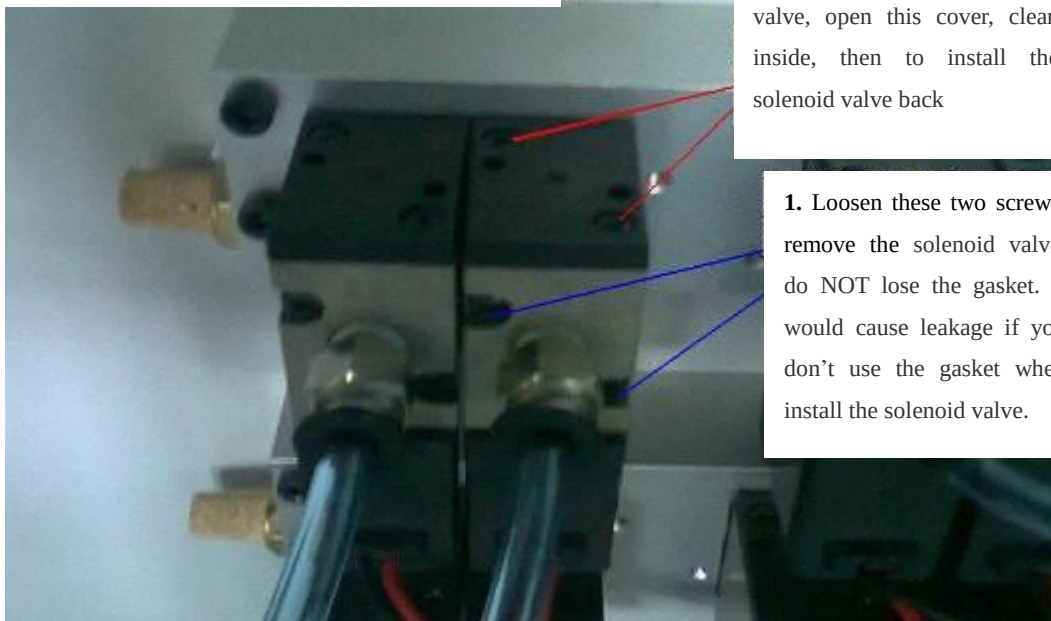
4-1. Main adjusted method for the spacing of 12MM FEEDER



1. The spacing pieces in the oval shape groove of cylinder are movable. There are numbers engraved on both sides, one side is 4/8, the other side is 8/12. The meaning of these numbers are the spacing can be adjusted to 12*4/12*8 for one side and for the other side is 12*8/12*12.
2. Adjustment method: If you want to adjust to 12*4, loosen two pcs of Allen screw, adjust the spacing pieces, adjoin the left side of the oval shape groove (should be very closed and without any gap), the cylinder would move accordingly, and then lock the screws, press down the hand pressure connecting rod to the end, check if the gear is moved by one grid, if yes, the adjustment is successful. You can do similar if to adjust to 12*8: loose the screws, adjoin the right side of the oval shape groove, the lock the screws, check if the gear is moved by two grid. To adjust 12*12 is required to remove the spacing piece, turn it over, and put the side with number 12 and 8 as outer side, to lock the side with number 12.

4-2. Solution for the leakage of solenoid valve:

Find out the broken vent corresponding solenoid valve



2. After remove the solenoid valve, open this cover, clean inside, then to install the solenoid valve back

1. Loosen these two screws, remove the solenoid valve, do NOT lose the gasket. It would cause leakage if you don't use the gasket when install the solenoid valve.

4-3. FQA for the Industrial Personal Computer:

1. System is too slow or breakdown.
 - Solution:
 1. Install antivirus program to kill the viruses.
 2. Restore the operation system and kill the viruses.
 3. Replace the HDD.
2. Card driver installation
 - Can NOT find the moving card and image card.
3. Software installation

4-4. Questions for motor

There are 15 motor in total, there are 3 pieces of servo motor and 12 pieces of stepper motor. The fault phenomenon for motor is not working. If there is, we need to check the power supply to the motor driver whether is normal or not, if yes, then check the line whether is normal or not.

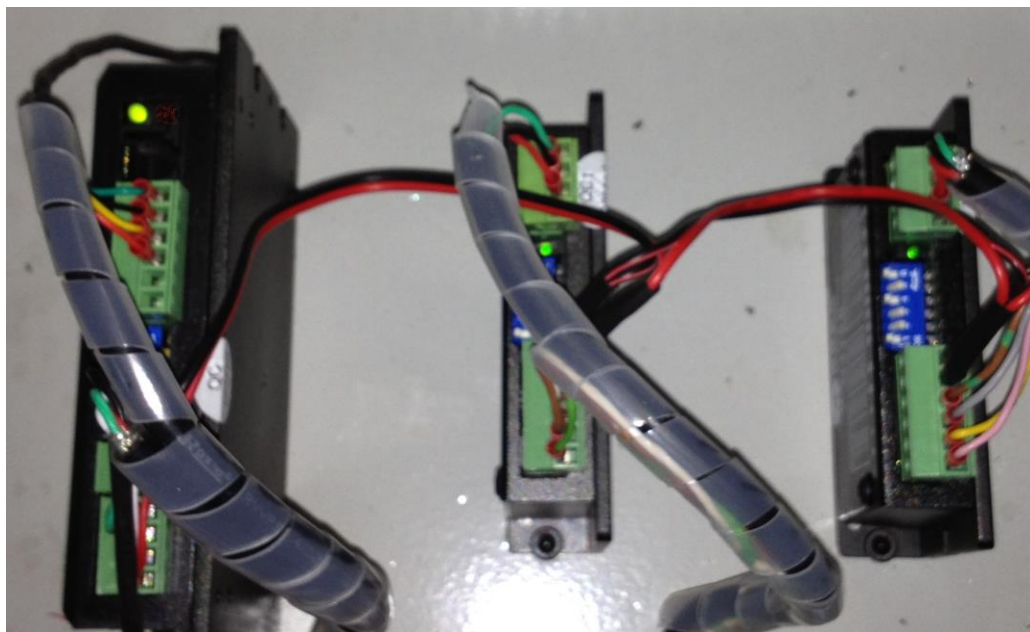


The driver of servo motor are in normal conditions.



The middle direction of the servo motor is in warning mode.

The probability that the servo motor and driver to be broken is very low. If there is a breakdown, we need to check are there something wrong with the line and the connector.



This picture shows the normal conditions of the stepper motor (Every green light of the driver is lighting.)

If the green light of the driver is not lighting, it might be the driver is broken or there is no power supply. If the green light of the driver is lighting, we need to check the line or maybe the motor is burnout.

4-5. Questions for suction nozzle when mounting:

1. Suction nozzle do not go down to the desired location.

- 1) the platform is not horizontal.
 - 2) the height of suction nozzle isn't setted appropriately.
2. Pick up the component after mounting.
- Reason: The vacuum pressure of the suction nozzle is too big , it causes the vacuum doesn't release fully, which lead to pick up the component again.
- Solution: Adjust the throttle of the vacuum pump to around -20.

4-6. Questions for suction nozzle when picking:

- 1.No vacuum for suction nozzle:
 - 1) The pressure of air source is too low
 - 2) Vacuum solenoid valve damage
 - 3) Vacuum filter block
 - 4) Vacuum pipe loose
 2. Suction height inaccurate
- Solution: Adjust the suction height of corresponding feeder in the parameter.
3. Suction position wrong
- Solution: Adjust the suction position of corresponding feeder in the parameter.

4-7. Placement accuracy setting method

1. Want place accurately, you need to make sure the picking is accurate. Placement would be accurate only if picking is accurate(picking inaccurate can be adjusted by its Suction position)
2. Several other important parameters
 - 1) head1 skews to X-direction of camera.
 - 2) head1 skews to Y-direction of camera.

Setting method: Located a special point on the platform , do NOT move the platform, use the camera to focus on this point, make it just in the center of the image, write down the coordinates of X, Y motor; then move the nozzle 1 to just right aiming at this special point on the platform, write down the coordinates of X, Y motor as well; so there would be two coordinates for X, Y motor, calculate their difference value which would be the distance of suction nozzle 1 of X and Y direction from the center of vidicon, input these distance into the above two parameters.

High parameters:

- 3) Suction nozzle distance in X direction:

This parameter is the distance of the center of two nozzle in X direction, its theoretical value is 36, considering installation error and gap, it could be approximately 36. We should follow up the reality distance.

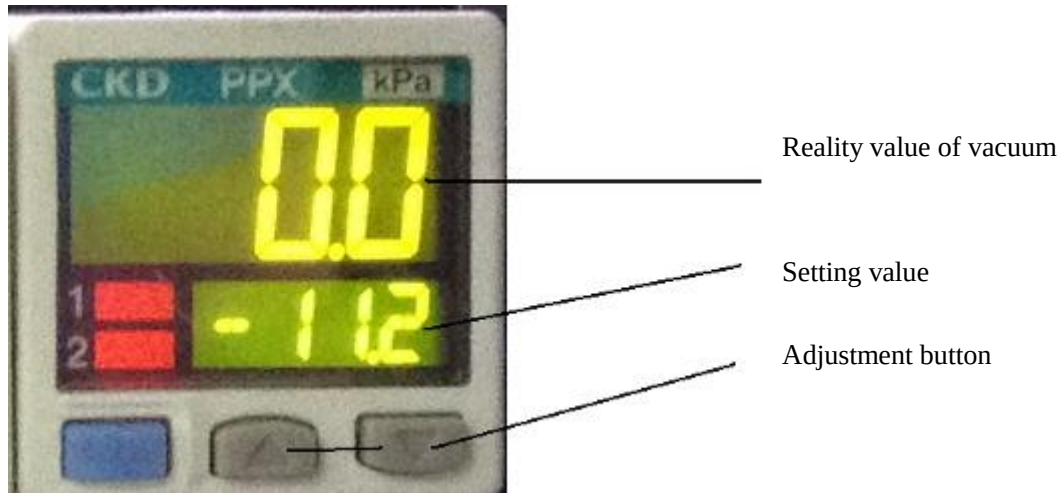
- 4) Suction nozzle distance in Y direction:

This parameter is the distance of the center of two nozzle in Y direction, its theoretical value is 0, considering installation error and gap, it could be approximately 0. We should follow up the reality distance.

3. Programming way would affect the accuracy

The aluminum PCB is the standard for programming, so make sure to put it horizontally. Put the aluminum PCB onto the platform, then move the camera to be above the aluminum PCB, move the platform, observe the image to see if the aluminum PCB is well set.

4-8. Vacuum pressure gage setting method



Vacuum gage

The reality vacuum value is shown as the above one. It can tell the vacuum of the suction nozzle after picking and without picking.

The setting value is shown as below one. If the machine during production detects the actual vacuum is smaller than the set value, it means the suction nozzle is picking the chips. If bigger, the picking is failed.

So the setting value should be set between value of picking chips and value of not-picking chips, then it would have a warning function (the setting value should be closed to value of not-picking).

Annex one: Daily maintenance list

		To avoid any accident, make sure to cut off the power supply before do the maintenance.						
Item			Frequency					
Method	Position	Confirmation way	Day	Week	Month	Every two months	Every half year	Year
Recondition	Air pressure	0.55MPa	○					
	Pipe and connector	Air leak		○				
	Starting indicator	on	○					
	Synchronous belt	Wear, damage、looseness		○				
	Rotating shaft belt	Wear, damage、looseness		○				
	Electric	Wear, damage		○				
Clean	Platform guide rail	Remove dust and other objects			○			
	Synchronous belt	Remove dust and other objects		○				
	Sensor	Every sensor			○			
	Suction nozzle	Clean suction nozzle		○				
	Suction nozzle outer shaft	Clean the inner shaft			○			
	Feeder station	Remove other objects		○				
	CCD camera	Remove dust and other objects		○				
	Display	Remove dust		○				
	Discharge platform	Remove dust and other objects	○					
	Upper shield	Remove dust		○				
Lubricate	X、Y guide rail	Lithium lubricating grease			○			
	Discharge platform guide rail	Lithium lubricating grease			○			

Annex 3: Suction nozzle list and its application component

Model no#	Application component
502	0402
503	0603, 0805
504	3014, 1208, 0805
505	3528, 5630, 5730
506	5050
507	1W